

AAE 50700: Principles of Dynamics Problem Set
2 Report

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(2.1) Rigid-body Kinematics and Dynamics

(a): Given

The following three matrices:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0.1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Find: Check whether each matrix is a valid DCM.

Analysis/Conclusions:

It is known that a Direction Cosine Matrix (DCM) must be both an orthogonal matrix and have a determinant equal to $\det = 1$ (a DCM could be valid with $\det = \pm 1$, but a -1 determinant would result in a reflection to a left-handed frame basis which is an improper frame basis) [1]. The former can be proven by showing that the candidate DCM multiplied by its transpose is equal to the identity matrix ($AA^T = I$), and the latter can of course just be proved by taking the determinant.

We see the results of these tests on each matrix below. The determinants were found using the `det()` function in Matlab:

We see that A **passes** both tests and **is a DCM**.

$$AA^T = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \quad (2)$$
$$\det(A) = 1$$

We see that B **fails** the $\det = 1$ requirement and **is NOT a DCM**.

$$BB^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \quad (3)$$
$$\det(B) = -1$$

We see that C **fails** the orthogonality requirement ($CC^T = I$) and **is NOT a DCM**.

$$CC^T = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0.1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0.1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.1 & 0 \\ 0.1 & 1.1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \neq I \quad (4)$$
$$\det(C) = 1$$

(b): Given

Consider a principal rotation of angle $\theta = 120 \text{ deg}$ about the axis $\hat{\lambda} = \frac{1}{\sqrt{3}}[1, 1, 1]^T$.

(b.1): Find

Compute the corresponding Euler parameters ϵ . Confirm $\|\epsilon\| = 1$.

Analysis/Conclusions: An Euler parameter vector ϵ can be found simply using:

$$\epsilon = \begin{bmatrix} \hat{\lambda} \sin(\theta/2) \\ \cos(\theta/2) \end{bmatrix} \in \mathbb{R}^4 \quad (5)$$

Thus we can compute this Euler parameter as:

$$\epsilon = \begin{bmatrix} \hat{\lambda} \sin(\theta/2) \\ \cos(\theta/2) \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} \sin(120 \text{ deg} / 2) \\ \frac{1}{\sqrt{3}} \sin(120 \text{ deg} / 2) \\ \frac{1}{\sqrt{3}} \sin(120 \text{ deg} / 2) \\ \cos(120 \text{ deg} / 2) \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} \quad (6)$$

Then to verify $\|\epsilon\| = 1$:

$$\|\epsilon\| = \sqrt{0.5^2 + 0.5^2 + 0.5^2 + 0.5^2} = 1 \quad (7)$$

(b.2): Find

Compute the corresponding DCM $[BN]$. Verify it is a valid DCM.

Analysis/Conclusions: An inertial frame to body frame DCM $[BN]$ is defined from Euler parameters as:

$$[BN] = \begin{bmatrix} 1 - 2\epsilon_2^2 - 2\epsilon_3^2 & 2(\epsilon_1\epsilon_2 + \epsilon_3\epsilon_4) & 2(\epsilon_1\epsilon_3 - \epsilon_2\epsilon_4) \\ 2(\epsilon_1\epsilon_2 - \epsilon_3\epsilon_4) & 1 - 2\epsilon_1^2 - 2\epsilon_3^2 & 2(\epsilon_2\epsilon_3 + \epsilon_1\epsilon_4) \\ 2(\epsilon_1\epsilon_3 + \epsilon_2\epsilon_4) & 2(\epsilon_2\epsilon_3 - \epsilon_1\epsilon_4) & 1 - 2\epsilon_1^2 - 2\epsilon_2^2 \end{bmatrix} \quad (8)$$

We then simply plug in the Euler parameter vector from (b.1):

$$\begin{aligned}
 [BN] &= \begin{bmatrix} 1 - 2(0.5)^2 - 2(0.5)^2 & 2((0.5)(0.5) + (0.5)(0.5)) & 2((0.5)(0.5) - (0.5)(0.5)) \\ 2((0.5)(0.5) - (0.5)(0.5)) & 1 - 2(0.5)^2 - 2(0.5)^2 & 2((0.5)(0.5) + (0.5)(0.5)) \\ 2((0.5)(0.5) + (0.5)(0.5)) & 2((0.5)(0.5) - (0.5)(0.5)) & 1 - 2(0.5)^2 - 2(0.5)^2 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}
 \end{aligned} \tag{9}$$

Then we can run the same tests as done in part (a) to verify the DCM is valid. So, the DCM must be orthogonal (proved by $[BN][BN]^T = I$) and must have a determinant of $\det = 1$ which is once again computed using Matlab $\det()$ function.

We see that $[BN]$ **passes** both tests and **is a DCM**.

$$\begin{aligned}
 [BN][BN]^T &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \\
 \det([BN]) &= 1
 \end{aligned} \tag{10}$$

(b.3): Find

Recompute $[BN]$ using $-\epsilon$ instead of ϵ , and confirm I get the same DCM.

Analysis/Conclusions: The Euler parameter $-\epsilon$ would simply be:

$$-\epsilon = - \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -0.5 \\ -0.5 \\ -0.5 \\ -0.5 \end{bmatrix} \tag{11}$$

Thus when we plug the values of this negated Euler parameter $-\epsilon$ into the DCM in equation (8), we get:

$$\begin{aligned}
 &= \begin{bmatrix} 1 - 2(-0.5)^2 - 2(-0.5)^2 & 2((-0.5)(-0.5) + (-0.5)(-0.5)) & 2((-0.5)(-0.5) - (-0.5)(-0.5)) \\ 2((-0.5)(-0.5) - (-0.5)(-0.5)) & 1 - 2(-0.5)^2 - 2(-0.5)^2 & 2((-0.5)(-0.5) + (-0.5)(-0.5)) \\ 2((-0.5)(-0.5) + (-0.5)(-0.5)) & 2((-0.5)(-0.5) - (-0.5)(-0.5)) & 1 - 2(-0.5)^2 - 2(-0.5)^2 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = [BN]
 \end{aligned} \tag{12}$$

Which is an identical DCM to part (b.2).

(c): Given

The starting point $\dot{\mathbf{H}}_c = \boldsymbol{\tau}_c$, where $\mathbf{H}_c$ is the angular momentum of the rigid body about its center of mass, $\boldsymbol{\tau}_c$ is the net external torque about the same point, and the overdot denotes differentiation with respect to time in an inertial frame.

Define Reference Frames:

- $\mathcal{N}:\{\mathcal{O},\hat{n}_1,\hat{n}_2,\hat{n}_3\}$: **Inertial reference frame**
A generic inertial reference frame with its origin at $\mathcal{O}$.
- $\mathcal{B}:\{\mathcal{C},\hat{b}_1,\hat{b}_2,\hat{b}_3\}$: **Body-fixed reference frame**
A frame fixed in the rigid body with its origin at the body's center of mass $\mathcal{C}$, rotating with respect to $\mathcal{N}$ at $\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}}$ (angular rate).

(c.1): Find

Show the Euler's rotational equations of motion $I_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega}) = \boldsymbol{\tau}_c$. In the process, justify why:

- $\frac{{}^{\mathcal{B}}d}{dt}\boldsymbol{\omega} = \dot{\boldsymbol{\omega}}$
- $\frac{{}^{\mathcal{B}}d}{dt}I_c = \mathbf{0}$

Where I_c is the inertia tensor of the rigid body and $\boldsymbol{\omega}$ is the angular rate vector of the rigid body with respect to some inertial frame. For the sake of notation, we will consider:

$$\boldsymbol{\omega} = \boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} \quad (13)$$

Analysis/Conclusions: To show that $I_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega}) = \boldsymbol{\tau}_c$, we begin from the given starting point:

$$\boldsymbol{\tau}_c = \dot{\mathbf{H}}_c \quad (14)$$

The angular momentum derivative $\dot{\mathbf{H}}_c$ can also be represented by the transport theorem utilizing $\mathcal{B}$ -frame relative parameters:

$$\dot{\mathbf{H}}_c = \frac{{}^{\mathcal{B}}d}{dt}\mathbf{H}_c + \boldsymbol{\omega} \times \mathbf{H}_c \quad (15)$$

Which substitutes to our starting point:

$$\boldsymbol{\tau}_c = \frac{{}^{\mathcal{B}}d}{dt}\mathbf{H}_c + \boldsymbol{\omega} \times \mathbf{H}_c \quad (16)$$

It is known that, by definition, the angular momentum relates to the inertia tensor and the angular rate vector as:

$$\mathbf{H}_c = I_c\boldsymbol{\omega} \quad (17)$$

Which can then be further substituted:

$$\begin{aligned}
\boldsymbol{\tau}_c &= \frac{{}^{\mathcal{B}}d}{dt} \mathbf{H}_c + \boldsymbol{\omega} \times \mathbf{H}_c \\
&= \frac{{}^{\mathcal{B}}d}{dt} (I_c \boldsymbol{\omega}) + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\
&= \frac{{}^{\mathcal{B}}dI_c}{dt} \boldsymbol{\omega} + I_c \frac{{}^{\mathcal{B}}d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega})
\end{aligned} \tag{18}$$

The above expression leaves us with the two terms the problem requested justification to the solutions of: $\frac{{}^{\mathcal{B}}dI_c}{dt}$ and $\frac{{}^{\mathcal{B}}d\boldsymbol{\omega}}{dt}$.

From the problem definition, we know that it is expected that:

$$\frac{{}^{\mathcal{B}}dI_c}{dt} = 0 \tag{19}$$

Which is conceptually simple to justify. Since the inertial tensor represents the moments and products of inertia of the rigid body, the values of the terms would shift as the shape of the rigid body changes orientation in a frame. Since the body frame $\mathcal{B}$ is by definition locked in orientation to the rigid body, the terms of the inertia tensor should never change with respect to the body, and thus its $\mathcal{B}$ -frame relative derivative should always be zero.

The other term's value given by the problem is

$$\frac{{}^{\mathcal{B}}d\boldsymbol{\omega}}{dt} = \dot{\boldsymbol{\omega}} \tag{20}$$

This can be quickly proved mathematically by applying the transport theorem to the inertial derivative of $\boldsymbol{\omega}$:

$$\begin{aligned}
\dot{\boldsymbol{\omega}} &= \frac{{}^{\mathcal{B}}d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega} \times \boldsymbol{\omega} \\
\dot{\boldsymbol{\omega}} &= \frac{{}^{\mathcal{B}}d\boldsymbol{\omega}}{dt} + \mathbf{0} \\
\dot{\boldsymbol{\omega}} &= \frac{{}^{\mathcal{B}}d\boldsymbol{\omega}}{dt}
\end{aligned} \tag{21}$$

As can be seen, the cross product term of $\boldsymbol{\omega} \times \boldsymbol{\omega}$ zeros out and we end up with the given term.

Then with these two terms justified, we can complete the proof for the Euler's equations of rotational motion:

$$\begin{aligned}
\boldsymbol{\tau}_c &= \frac{{}^{\mathcal{B}}dI_c}{dt}\boldsymbol{\omega} + I_c \frac{{}^{\mathcal{B}}d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega}) \\
\boldsymbol{\tau}_c &= \mathbf{0}\boldsymbol{\omega} + I_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega}) \\
\boldsymbol{\tau}_c &= I_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega})
\end{aligned} \tag{22}$$

(c.2): Find

Assuming $\mathcal{B}$ -frame axes are principal axes of inertia, derive the three scalar differential equations from $\boldsymbol{\tau}_c = I_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega})$.

Analysis/Conclusions: When we assume the $\mathcal{B}$ -frame axes are principal axes of inertia, the inertia tensor is equivalent to a diagonal:

$$I_c = \text{diag}(I_1, I_2, I_3) \tag{23}$$

So the scalar differential equations can be simply derived from the Euler equations of rotational motion:

$$\begin{aligned}
\boldsymbol{\tau}_c &= I_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega}) \\
\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} &= \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \times \left(\begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \right) \\
\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} &= \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \times \begin{bmatrix} I_1\omega_1 \\ I_2\omega_2 \\ I_3\omega_3 \end{bmatrix} \\
\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} &= \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} I_3\omega_2\omega_3 - I_2\omega_2\omega_3 \\ I_1\omega_1\omega_3 - I_3\omega_1\omega_3 \\ I_2\omega_1\omega_2 - I_1\omega_1\omega_2 \end{bmatrix} \\
\begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} &= \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} - \begin{bmatrix} I_3\omega_2\omega_3 - I_2\omega_2\omega_3 \\ I_1\omega_1\omega_3 - I_3\omega_1\omega_3 \\ I_2\omega_1\omega_2 - I_1\omega_1\omega_2 \end{bmatrix} \\
\begin{bmatrix} I_1\dot{\omega}_1 \\ I_2\dot{\omega}_2 \\ I_3\dot{\omega}_3 \end{bmatrix} &= \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} - \begin{bmatrix} I_3\omega_2\omega_3 - I_2\omega_2\omega_3 \\ I_1\omega_1\omega_3 - I_3\omega_1\omega_3 \\ I_2\omega_1\omega_2 - I_1\omega_1\omega_2 \end{bmatrix} \\
\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} &= \begin{bmatrix} I_1\dot{\omega}_1 \\ I_2\dot{\omega}_2 \\ I_3\dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} I_3\omega_2\omega_3 - I_2\omega_2\omega_3 \\ I_1\omega_1\omega_3 - I_3\omega_1\omega_3 \\ I_2\omega_1\omega_2 - I_1\omega_1\omega_2 \end{bmatrix} \\
\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} &= \begin{bmatrix} I_1\dot{\omega}_1 \\ I_2\dot{\omega}_2 \\ I_3\dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} (I_3 - I_2)\omega_2\omega_3 \\ (I_1 - I_3)\omega_1\omega_3 \\ (I_2 - I_1)\omega_1\omega_2 \end{bmatrix}
\end{aligned} \tag{24}$$

Thus the three scalar differential equations from the Euler equations of rotational motion are:

$$\begin{aligned}\tau_1 &= I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_3 \omega_2 \\ \tau_2 &= I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_3 \omega_1 \\ \tau_3 &= I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2\end{aligned}\tag{25}$$

(d): Given

Consider torque-free motion of an axisymmetric rigid body, $I_1 = I_2 = I_T$ and $I_3 = J$ (with $J \neq I_T$), using Euler's equations from (c).

(d.1): Find

Show that $\omega_3(t) = \omega_{3_0}$ is constant and that ω_1 and ω_2 satisfy a pair of coupled linear ODEs. (ω_{3_0} represents the initial angular rate of ω_3)

Analysis/Conclusions: First we must substitute the moment of inertia terms given by the problem definition into the scalar differential equations found in (c) from Euler's equations of rotational motion. We also zero out all torque terms because this is a torque-free motion scenario:

$$\begin{aligned}0 &= I_T \dot{\omega}_1 + (J - I_T) \omega_3 \omega_2 \\ 0 &= I_T \dot{\omega}_2 + (I_T - J) \omega_3 \omega_1 \\ 0 &= J \dot{\omega}_3 + (I_T - I_T) \omega_1 \omega_2\end{aligned}\tag{26}$$

Then the steps to prove that $\omega_3(t) = \omega_{3_0}$ (constant ω_3) are fairly simple:

$$\begin{aligned}0 &= J \dot{\omega}_3 + (I_T - I_T) \omega_1 \omega_2 \\ 0 &= J \dot{\omega}_3 + 0 \omega_1 \omega_2 \\ \frac{0}{J} &= \dot{\omega}_3 \\ 0 &= \dot{\omega}_3\end{aligned}\tag{27}$$

Since $\dot{\omega}_3 = 0$ for all t , then $\omega_3(t)$ must always equal its initial value ω_{3_0} .

$$\omega_3(t) = \omega_{3_0}\tag{28}$$

Then to show that ω_1 and ω_2 satisfy a pair of coupled linear ODE's, we first

rewrite the other two equations in (26) in simpler terms:

$$\begin{aligned}
\text{Let: } K &= \frac{I_T - J}{I_T} \\
0 &= I_T \dot{\omega}_1 + (J - I_T) \omega_3 \omega_2 \\
I_T \dot{\omega}_1 &= (I_T - J) \omega_3 \omega_2 \\
\dot{\omega}_1 &= \frac{I_T - J}{I_T} \omega_3 \omega_2 \\
\dot{\omega}_1 &= K \omega_{3_0} \omega_2(t) \\
0 &= I_T \dot{\omega}_2 + (I_T - J) \omega_3 \omega_1 \\
I_T \dot{\omega}_2 &= -(I_T - J) \omega_3 \omega_1 \\
\dot{\omega}_2 &= -K \omega_{3_0} \omega_1(t)
\end{aligned} \tag{29}$$

Thus we see from the system of equations:

$$\begin{aligned}
\dot{\omega}_1 &= K \omega_{3_0} \omega_2(t) \\
\dot{\omega}_2 &= -K \omega_{3_0} \omega_1(t)
\end{aligned} \tag{30}$$

That this is a linear system of equations because each equation is a single function multiplied by a constant ($K \omega_{3_0}$). We also see that $\dot{\omega}_1(t)$ is dependent on $\omega_2(t)$ and $\dot{\omega}_2(t)$ is dependent on $\omega_1(t)$. This makes ω_1 and ω_2 part of a pair of coupled linear ODE's.

(d.2): Find

Solve the coupled linear ODEs to show that, with initial condition $\boldsymbol{\omega}_0(t) = [\omega_{1_0}, \omega_{2_0}, \omega_{3_0}]^T$,

$$\begin{aligned}
\omega_1(t) &= \omega_{1_0} \cos(K \omega_{3_0} t) + \omega_{2_0} \sin(K \omega_{3_0} t) \\
\omega_2(t) &= \omega_{2_0} \cos(K \omega_{3_0} t) - \omega_{1_0} \sin(K \omega_{3_0} t)
\end{aligned} \tag{31}$$

Where: $K = (I_T - J)/I_T$

Analysis/Conclusions: The two equations found in (d.1) are coupled, so neither can be integrated on its own. However, since the coefficient $K \omega_{3_0}$ was shown to be constant, the pair can be decoupled by differentiating one of them and substituting the other in. To shorten the expressions, let $\lambda = K \omega_{3_0}$, so the system from (d.1) is written as:

$$\begin{aligned}
\dot{\omega}_1 &= \lambda \omega_2 \\
\dot{\omega}_2 &= -\lambda \omega_1
\end{aligned} \tag{32}$$

Differentiating the first equation with respect to time and substituting the second one in for $\dot{\omega}_2$ removes ω_2 entirely:

$$\begin{aligned}\ddot{\omega}_1 &= \lambda \dot{\omega}_2 \\ &= \lambda (-\lambda \omega_1) \\ \ddot{\omega}_1 + \lambda^2 \omega_1 &= 0\end{aligned}\tag{33}$$

This is the standard undamped harmonic oscillator equation, whose general solution is a sinusoid of frequency λ with two unknown coefficients:

$$\omega_1(t) = A \cos(\lambda t) + B \sin(\lambda t)\tag{34}$$

The two coefficients are found from the initial conditions. Evaluating $\omega_1(t)$ at $t = 0$ immediately gives A :

$$\omega_1(0) = A \cos(0) + B \sin(0) = A = \omega_{1_0}\tag{35}$$

Then B is found by differentiating the general solution and evaluating it at $t = 0$, where the value of $\dot{\omega}_1(0)$ comes from the first ODE of the system:

$$\begin{aligned}\dot{\omega}_1(t) &= -A\lambda \sin(\lambda t) + B\lambda \cos(\lambda t) \\ \dot{\omega}_1(0) &= B\lambda \\ \dot{\omega}_1(0) &= \lambda \omega_2(0) = \lambda \omega_{2_0} \\ B\lambda &= \lambda \omega_{2_0} \\ B &= \omega_{2_0}\end{aligned}\tag{36}$$

With both coefficients known, $\omega_2(t)$ follows without solving a second differential equation, since the first ODE already relates it directly to $\dot{\omega}_1$:

$$\begin{aligned}\omega_2(t) &= \frac{\dot{\omega}_1(t)}{\lambda} \\ &= \frac{-\omega_{1_0} \lambda \sin(\lambda t) + \omega_{2_0} \lambda \cos(\lambda t)}{\lambda} \\ &= \omega_{2_0} \cos(\lambda t) - \omega_{1_0} \sin(\lambda t)\end{aligned}\tag{37}$$

Substituting $\lambda = K\omega_{3_0}$ back in gives the solutions stated in the problem definition, which completes the derivation:

$$\begin{aligned}\omega_1(t) &= \omega_{1_0} \cos(K\omega_{3_0} t) + \omega_{2_0} \sin(K\omega_{3_0} t) \\ \omega_2(t) &= \omega_{2_0} \cos(K\omega_{3_0} t) - \omega_{1_0} \sin(K\omega_{3_0} t) \\ \omega_3(t) &= \omega_{3_0}\end{aligned}\tag{38}$$

(2.2) Spacecraft Attitude Dynamics

(e):

Define Reference Frames:

- $\mathcal{N}:\{\mathcal{O},\hat{n}_1,\hat{n}_2,\hat{n}_3\}$: **Inertial reference frame**
A generic inertial reference frame with its origin at $\mathcal{O}$.
- $\mathcal{B}:\{\mathcal{C},\hat{b}_1,\hat{b}_2,\hat{b}_3\}$: **Body-fixed reference frame**
A frame fixed in the rigid body with its origin at the body's center of mass $\mathcal{C}$, rotating with respect to $\mathcal{N}$ at $\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}}$ (angular rate).

Given: Starting from the (approximate) gravity-gradient torque about the center of mass $\boldsymbol{\tau}_G = \frac{3\mu}{r_c^5} \mathbf{r}_c \times (I_c \mathbf{r}_c)$.

Find: Derive the component form in $\mathcal{B}$ -frame:

$$\tau_1 = \frac{3\mu}{r_c^5} (I_3 - I_2) r_{c3} r_{c2}, \quad \tau_2 = \frac{3\mu}{r_c^5} (I_1 - I_3) r_{c1} r_{c3}, \quad \tau_3 = \frac{3\mu}{r_c^5} (I_2 - I_1) r_{c2} r_{c1} \quad (39)$$

where we assume the body-frame ($\mathcal{B}$) axes are principal axes of inertia, and ${}^{\mathcal{B}}\mathbf{r}_c = [r_{c1}, r_{c2}, r_{c3}]^T$.

In this problem we define $\mathbf{r}_c$ as the spacecraft center of mass position from the attractive center of the body it orbits, r_{ci} as the position components, μ as the gravitational parameter of the orbited body, and I_i as the moment of inertia on different axes like in (c) and (d).

Analysis/Conclusions: Just like in part (c) since the $\mathcal{B}$ -frame axes are the principal axes of inertia, we can represent the inertia tensor as:

$$I_c = \text{diag}(I_1, I_2, I_3) \quad (40)$$

Then with this we can substitute terms into the gravity-gradient torque approximation. Since this is all done with $\mathcal{B}$ -frame as the principal axes of inertia, all

matrix form vectors are representing $\mathcal{B}$ -frame vectors:

$$\begin{aligned}
\boldsymbol{\tau}_G &= \frac{3\mu}{r_c^5} \mathbf{r}_c \times (I_c \mathbf{r}_c) \\
\boldsymbol{\tau}_G &= \frac{3\mu}{r_c^5} \begin{bmatrix} r_{c1} \\ r_{c2} \\ r_{c3} \end{bmatrix} \times \left(\begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} r_{c1} \\ r_{c2} \\ r_{c3} \end{bmatrix} \right) \\
\boldsymbol{\tau}_G &= \frac{3\mu}{r_c^5} \begin{bmatrix} r_{c1} \\ r_{c2} \\ r_{c3} \end{bmatrix} \times \begin{bmatrix} I_1 r_{c1} \\ I_2 r_{c2} \\ I_3 r_{c3} \end{bmatrix} \\
\boldsymbol{\tau}_G &= \frac{3\mu}{r_c^5} \begin{bmatrix} (I_3 - I_2) r_{c2} r_{c3} \\ (I_1 - I_3) r_{c1} r_{c3} \\ (I_2 - I_1) r_{c1} r_{c2} \end{bmatrix} \\
\boldsymbol{\tau}_G &= \frac{3\mu}{r_c^5} \left((I_3 - I_2) r_{c2} r_{c3} \hat{\mathbf{b}}_1 + (I_1 - I_3) r_{c1} r_{c3} \hat{\mathbf{b}}_2 + (I_2 - I_1) r_{c1} r_{c2} \hat{\mathbf{b}}_3 \right)
\end{aligned} \tag{41}$$

Then using the $\mathcal{B}$ -frame component form of $\boldsymbol{\tau}_G = \tau_1 \hat{\mathbf{b}}_1 + \tau_2 \hat{\mathbf{b}}_2 + \tau_3 \hat{\mathbf{b}}_3$ we get the given equations in the problem definition:

$$\tau_1 \hat{\mathbf{b}}_1 + \tau_2 \hat{\mathbf{b}}_2 + \tau_3 \hat{\mathbf{b}}_3 = \frac{3\mu}{r_c^5} \left((I_3 - I_2) r_{c2} r_{c3} \hat{\mathbf{b}}_1 + (I_1 - I_3) r_{c1} r_{c3} \hat{\mathbf{b}}_2 + (I_2 - I_1) r_{c1} r_{c2} \hat{\mathbf{b}}_3 \right) \tag{42}$$

Splitting the equations out per component:

$$\begin{aligned}
\tau_1 \hat{\mathbf{b}}_1 &= \frac{3\mu}{r_c^5} (I_3 - I_2) r_{c2} r_{c3} \hat{\mathbf{b}}_1 \\
\tau_1 &= \frac{3\mu}{r_c^5} (I_3 - I_2) r_{c2} r_{c3} \\
\tau_2 \hat{\mathbf{b}}_2 &= \frac{3\mu}{r_c^5} (I_1 - I_3) r_{c1} r_{c3} \hat{\mathbf{b}}_2 \\
\tau_2 &= \frac{3\mu}{r_c^5} (I_1 - I_3) r_{c1} r_{c3} \\
\tau_3 \hat{\mathbf{b}}_3 &= \frac{3\mu}{r_c^5} (I_2 - I_1) r_{c1} r_{c2} \hat{\mathbf{b}}_3 \\
\tau_3 &= \frac{3\mu}{r_c^5} (I_2 - I_1) r_{c1} r_{c2}
\end{aligned} \tag{43}$$

Which completes the derivation.

(f): Given

Consider an axisymmetric rigid body with $I_1 = I_2 = I_T = 10 \text{ kg} \cdot \text{m}^2$, $I_3 = J = 5 \text{ kg} \cdot \text{m}^2$, in torque-free motion with initial angular velocity $\boldsymbol{\omega}_0 = [0.2, 0.1, 1.0]^T \text{ rad/s}$.

Define Reference Frames:

- $\mathcal{N}:\{\mathcal{O}, \hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2, \hat{\mathbf{n}}_3\}$: **Inertial reference frame**
A generic inertial reference frame with its origin at $\mathcal{O}$.
- $\mathcal{B}:\{\mathcal{C}, \hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3\}$: **Body-fixed reference frame**
A frame fixed in the rigid body with its origin at the body's center of mass $\mathcal{C}$, rotating with respect to $\mathcal{N}$ at $\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}}$ (angular rate).

(f.1) Find

Numerically integrate Euler's equations of motion for at least 120 sec. Report relevant plots.

Analysis/Conclusions: The first step is to rewrite Euler's equations of rotation motion in a form that can be numerically integrated by putting the $\dot{\boldsymbol{\omega}}$ term on its own on one side of the equation. Since this is torque-free motion, the torque term zeros out:

$$\begin{aligned}\boldsymbol{\tau}_c &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\ I_c \dot{\boldsymbol{\omega}} &= \boldsymbol{\tau}_c - \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\ \dot{\boldsymbol{\omega}} &= I_c^{-1} (\boldsymbol{\tau}_c - \boldsymbol{\omega} \times (I_c \boldsymbol{\omega})) \\ \dot{\boldsymbol{\omega}} &= I_c^{-1} (\mathbf{0} - \boldsymbol{\omega} \times (I_c \boldsymbol{\omega})) \\ \dot{\boldsymbol{\omega}} &= I_c^{-1} (-\boldsymbol{\omega} \times (I_c \boldsymbol{\omega}))\end{aligned}\tag{44}$$

Then, the above equation is written into a function in Matlab with the constants given in the problem definition, and the function is numerically integrated for 120 seconds using ODE113. Below is the resulting plots of the angular rate on each axis in $\mathcal{B}$ -frame where $\boldsymbol{\omega} = \omega_1 \hat{\mathbf{b}}_1 + \omega_2 \hat{\mathbf{b}}_2 + \omega_3 \hat{\mathbf{b}}_3$:

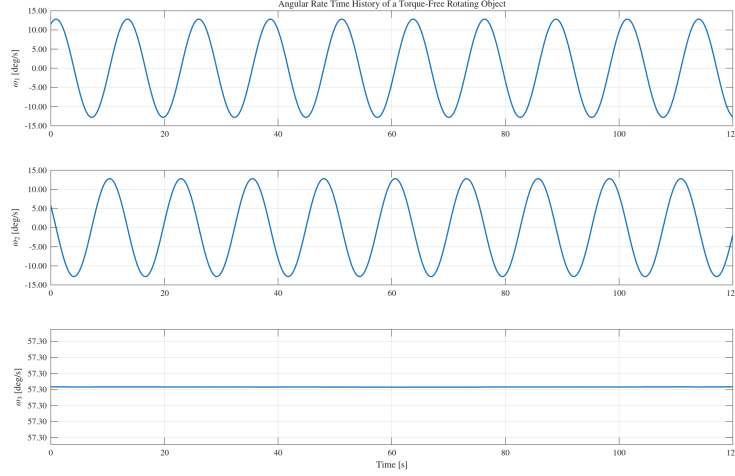


Figure 1: Time history of the angular rates ω_1 , ω_2 , and ω_3 .

Observe in the plots above that there is an oscillation in the angular rate of the $\hat{\mathbf{b}}_1$ and $\hat{\mathbf{b}}_2$ directions that appear to act like a sin or cos function approximately. Furthermore the $\hat{\mathbf{b}}_3$ axis angular rate stays constant to its initial condition. Both of these behaviors are consistent with analytic derivations of $\omega_1(t)$, $\omega_2(t)$, and $\omega_3(t)$ done in part (d).

(f.2) Find

Compute the value of $\boldsymbol{\omega}(t)$ using the analytical solution from part (d.2). Overlay the numerical and analytical solutions. The two should agree closely; discuss any discrepancy observed.

Analysis/Conclusions: To restate, the analytic expressions for $\boldsymbol{\omega}(t)$ stated in (d.2) were:

$$\begin{aligned}\omega_1(t) &= \omega_{1_0} \cos(K\omega_{3_0}t) + \omega_{2_0} \sin(K\omega_{3_0}t) \\ \omega_2(t) &= \omega_{2_0} \cos(K\omega_{3_0}t) - \omega_{1_0} \sin(K\omega_{3_0}t) \\ \omega_3(t) &= \omega_{3_0}\end{aligned}\tag{45}$$

Where: $K = (I_T - J)/I_T$

Extending the numerical integration code written for (f.2) to also plot the above analytic expressions, the following plots were made:

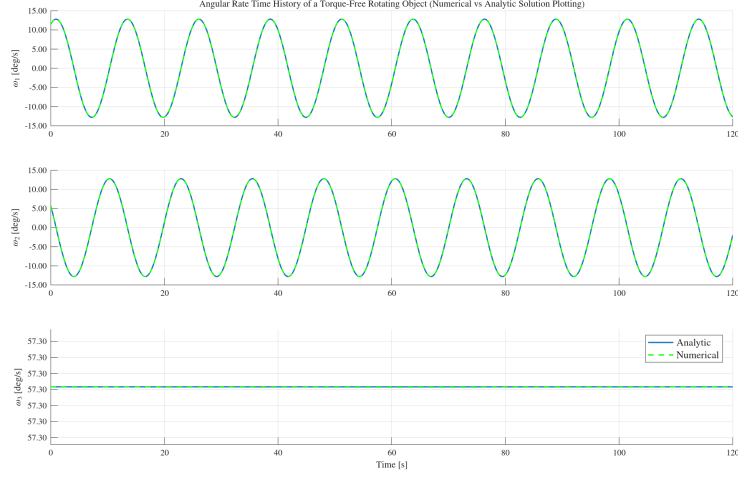


Figure 2: Time history of the angular rate ω comparing the analytic solution to a numerical integration of Euler's equations of rotational motion given the same parameters.

It can be seen that the two time histories of $\omega(t)$ from the analytic and numerical approaches follow each other closely.

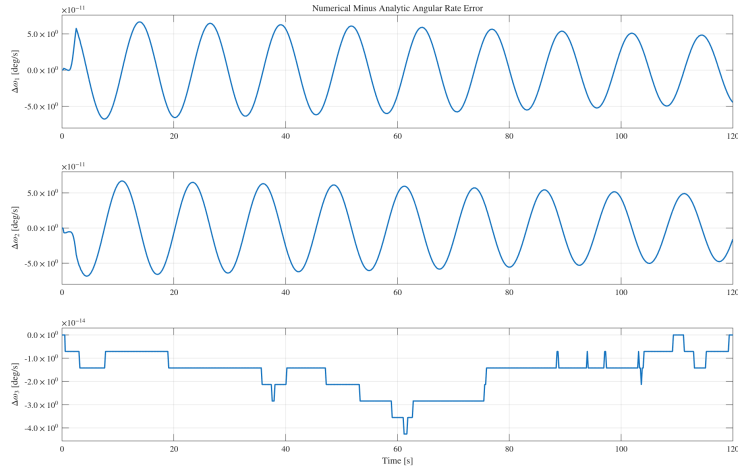


Figure 3: Plotting difference in $\omega_i(t)$ for all three directions between the analytic and numerical approaches to simulation.

When plotting the error of the two methods over time, it can be seen that

the difference of ω_3 is exceptionally small with some variance, which is expected because we are effectively comparing the small fluctuations in numerical drift to a true constant. The difference in ω_1 and ω_2 are both three orders of magnitude higher, but still an acceptably low order of magnitude (10^{-11}). It is interesting that once the numerical integration gets past the first few oscillations in ω_1 and ω_2 , the difference graphs themselves follow a similar oscillation pattern out of phase with the main graphs.

(g): Given

Consider a satellite made of a bus plus two identical solar panels.

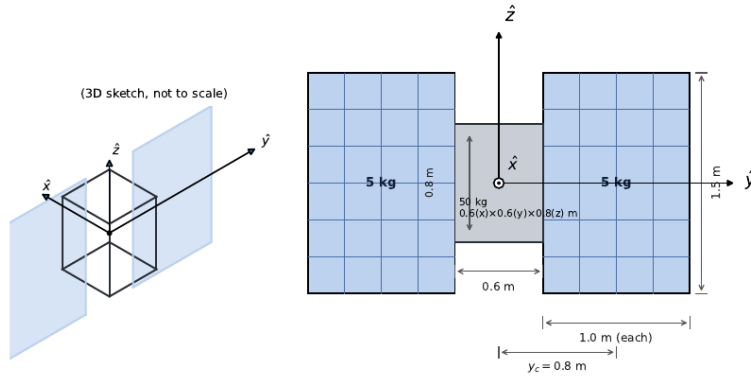


Figure 4: Bus-plus-panels satellite configuration.

- Bus: mass 50 kg , size $0.6\text{ m}(\hat{x}) \times 0.6\text{ m}(\hat{y}) \times 0.8\text{ m}(\hat{z})$, centered at the $\mathcal{B}$ -frame origin.
- Panels: two identical rectangular panels, $1.0\text{ m}(\text{along } \hat{y}) \times 1.5\text{ m}(\text{along } \hat{z})$, negligible thickness, 5 kg each, mounted symmetrically with their inner edge at $y = 0.3\text{ m}$ from the origin.
- The following equations can be used without derivation. For a box-shaped object of mass m with side lengths L_x, L_y, L_z , the moment of inertia about its own center of mass (CoM) is given by:

$$I_{xx} = \frac{m}{12} (L_y^2 + L_z^2), \quad I_{yy} = \frac{m}{12} (L_x^2 + L_z^2), \quad I_{zz} = \frac{m}{12} (L_x^2 + L_y^2) \quad (46)$$

Where m is the box's mass, I_i represents the moment of inertia component of that axis, and L_i is the length of that side of the box parallel to that axis.

Parallel Axis Theorem:

For a rigid body of mass m with inertia tensor I_c about its own center of mass,

its inertia tensor about a parallel reference point P displaced from the CoM by $\mathbf{d} \in \mathbb{R}^3$ is

$$I_P = I_c + m \left(\|\mathbf{d}\|^2 I_3 - \mathbf{d}\mathbf{d}^T \right) \quad (47)$$

For a composite body, sum each component's own-centroid inertia tensor plus its own parallel-axis correction to assemble the inertia tensor about one common point.

Define Reference Frames:

- $\mathcal{N}:\{\mathcal{O},\hat{n}_1,\hat{n}_2,\hat{n}_3\}$: **Inertial reference frame**
A generic inertial reference frame with its origin at $\mathcal{O}$.
- $\mathcal{B}:\{\mathcal{C},\hat{x},\hat{y},\hat{z}\}$: **Body-fixed reference frame**
A frame fixed in the rigid body with its origin at the body's center of mass $\mathcal{C}$, rotating with respect to $\mathcal{N}$ at $\omega_{\mathcal{B}/\mathcal{N}}$ (angular rate).

(g.1): Find

Explain why the system's CoM is exactly at the $\mathcal{B}$ -frame origin.

Analysis/Conclusions:

We know from the problem definition that the bus is centered at the $\mathcal{B}$ -frame origin. This means that depending on what other rigid bodies are affixed to the bus, the CoM of the entire combined body could be at different place than the $\mathcal{B}$ -frame origin.

It is known that the CoM of a system of multiple bodies is the sum of the CoM of each of the component bodies weighted by respective masses of the components. Likewise, when affixing a panel to the bus, the new center of mass between the two bodies will be somewhere on the line drawn between the original two CoM's. However, since we are affixing two identical panels symmetrically on either side of the bus, the three CoM's of the components can have a single line drawn through them. And since the equally massed panels are equidistant from the bus's CoM exactly at the center of that line, the displacement each panel would've caused in the unified CoM is canceled out by the other.

This means the CoM will still be where the bus's original CoM was and by extension the $\mathcal{B}$ -frame origin is still aligned with it.

(g.2): Find

Compute the inertia tensor of each component about the origin.

Analysis/Conclusions: In this problem, we once again assume that the $\mathcal{B}$ -frame axes are the principal axes of inertia. This means that the inertia tensor

of each component will be $\text{diag}(I_{xx}, I_{yy}, I_{zz})$ (which is a diagonal of the moment of inertia components) with the appropriate offset given by the parallel axis theorem. Below we compute each component's inertia tensor this way:

Given the parameters in the problem definition, the inertia tensor for the bus has the following moment of inertia components:

$$\begin{aligned}
I_{xx} &= \frac{m}{12} (L_y^2 + L_z^2) \\
&= \frac{50kg}{12} ((0.6m)^2 + (0.8m)^2) \\
&= 4.1666...kg \cdot m^2 \\
I_{yy} &= \frac{m}{12} (L_x^2 + L_z^2) \\
&= \frac{50kg}{12} ((0.6m)^2 + (0.8m)^2) \\
&= 4.1666...kg \cdot m^2 \\
I_{zz} &= \frac{m}{12} (L_x^2 + L_y^2) \\
&= \frac{50kg}{12} ((0.6m)^2 + (0.6m)^2) \\
&= 3kg \cdot m^2
\end{aligned} \tag{48}$$

Since the bus already has its geometric center and center of mass at the frame origin, there is no need to use the parallel axis theorem to compute an offset inertia tensor, and the bus's inertia tensor is:

$$I_{\text{bus}} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} = \begin{bmatrix} 4.1666... & 0 & 0 \\ 0 & 4.1666... & 0 \\ 0 & 0 & 3 \end{bmatrix} kg \cdot m^2 \tag{49}$$

Given the parameters in the problem definition, the inertia tensor for the both

panels have the following moment of inertia components:

$$\begin{aligned}
I_{xx} &= \frac{m}{12} (L_y^2 + L_z^2) \\
&= \frac{5kg}{12} ((1.0m)^2 + (1.5m)^2) \\
&= 1.3541...kg \cdot m^2 \\
I_{yy} &= \frac{m}{12} (L_x^2 + L_z^2) \\
&= \frac{5kg}{12} ((0m)^2 + (1.5m)^2) \\
&= 0.9375kg \cdot m^2 \\
I_{zz} &= \frac{m}{12} (L_x^2 + L_y^2) \\
&= \frac{5kg}{12} ((0m)^2 + (1.0m)^2) \\
&= 0.4166...kg \cdot m^2
\end{aligned} \tag{50}$$

Then the inertia tensor of both panels without the appropriate offset applied will be:

$$I_{\text{pan}} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} = \begin{bmatrix} 1.3541... & 0 & 0 \\ 0 & 0.9375 & 0 \\ 0 & 0 & 0.4166... \end{bmatrix} kg \cdot m^2 \tag{51}$$

Then we apply the offset for the first panel (which we will call the one in the positive $\hat{y}$ direction from the origin). Since the first panel is in the positive $\hat{y}$ direction, the displacement vector $\mathbf{d}$ is found by multiplying the distance from the bus's center of mass to the first panel's center of mass by the $\hat{y}$ basis vector. The distance is simply equal to half the $\hat{y}$ axis length of both objects added together:

$$\mathbf{d} = \frac{(L_{y_{\text{bus}}} + L_{y_{\text{panel1}}})}{2} \hat{\mathbf{y}} = \frac{(0.6m + 1.0m)}{2} \hat{\mathbf{y}} = 0.8m \hat{\mathbf{y}} \tag{52}$$

Then to find the true inertia tensor of the first panel by applying the appropriate offset by the parallel axis theorem:

$$\begin{aligned}
I_{\text{panel1}} &= I_{\text{pan}} + m (\|\mathbf{d}\|^2 I_3 - \mathbf{d} \mathbf{d}^T) \\
&= \begin{bmatrix} 1.3541... & 0 & 0 \\ 0 & 0.9375 & 0 \\ 0 & 0 & 0.4166... \end{bmatrix} kg \cdot m^2 + 5kg \left(\|0.8m \hat{\mathbf{y}}\|^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - (0.8m)^2 \hat{\mathbf{y}} \hat{\mathbf{y}}^T \right)
\end{aligned} \tag{53}$$

Which then in Matlab compute together as:

$$I_{\text{panel1}} = \begin{bmatrix} 4.5541... & 0 & 0 \\ 0 & 0.9375 & 0 \\ 0 & 0 & 3.6166... \end{bmatrix} kg \cdot m^2 \quad (54)$$

Which is the first panel's component inertia tensor about the origin.

Then to find the second panel, since it is in the exact same location mirrored about the $\hat{z}$ axis, the exact same steps are repeated but with the $\mathbf{d}$ vector negated:

$$\begin{aligned} I_{\text{panel2}} &= I_{\text{pan}} + m (\|\mathbf{d}\|^2 I_3 - \mathbf{d}\mathbf{d}^T) \\ &= \begin{bmatrix} 1.3541... & 0 & 0 \\ 0 & 0.9375 & 0 \\ 0 & 0 & 0.4166... \end{bmatrix} kg \cdot m^2 + 5kg \left(\|\mathbf{d}\|^2 I_3 - \mathbf{d}\mathbf{d}^T \right) \end{aligned} \quad (55)$$

Which then in Matlab compute together as:

$$I_{\text{panel2}} = \begin{bmatrix} 4.5541... & 0 & 0 \\ 0 & 0.9375 & 0 \\ 0 & 0 & 3.6166... \end{bmatrix} kg \cdot m^2 \quad (56)$$

Which, as can be seen, results in an identical tensor about the origin to that of the first panel.

(g.3): Find

Compute the assembled inertia tensor I_c about the system origin by summing the three contributions. Verify that very product of inertia (I_{xy}, I_{xz}, I_{yz}) is exactly zero.

Analysis/Conclusions: As stated in the problem, the assembled inertia tensor I_c is simply the sum of the three contributions:

$$\begin{aligned} I_c &= I_{\text{bus}} + I_{\text{panel1}} + I_{\text{panel2}} \\ &= \left(\begin{bmatrix} 4.1666... & 0 & 0 \\ 0 & 4.1666... & 0 \\ 0 & 0 & 3 \end{bmatrix} + \begin{bmatrix} 4.5541... & 0 & 0 \\ 0 & 0.9375 & 0 \\ 0 & 0 & 3.6166... \end{bmatrix} + \begin{bmatrix} 4.5541... & 0 & 0 \\ 0 & 0.9375 & 0 \\ 0 & 0 & 3.6166... \end{bmatrix} \right) kg \cdot m^2 \end{aligned} \quad (57)$$

Which are summed together in Matlab as:

$$I_c = \begin{bmatrix} 13.2750... & 0 & 0 \\ 0 & 6.0416... & 0 \\ 0 & 0 & 10.2333... \end{bmatrix} kg \cdot m^2 \quad (58)$$

Which makes the assembled tensor I_c .

The values in the above matrix are copied exactly from the output of my Matlab code. Likewise, it can be seen that **all product of inertia** terms (non-diagonal terms) are **exactly zero**.

(h): Given

Consider the satellite in part (g) in the orbit characterized in the following table:

Table 1: Reference orbit

Constants			Classical orbital elements at epoch			
μ_{earth}	R_{earth}	J_2	a	e	i	$\Omega_0, \omega_0, \nu_0$
398600 km ³ /s ²	6378.137 km	1.08263×10^{-3}	6978 km	0	45 deg	0 deg

Define Reference Frames:

- $\mathcal{N}:\{\mathcal{O}, \hat{n}_1, \hat{n}_2, \hat{n}_3\}$: **Inertial reference frame**
A generic inertial reference frame with its origin at $\mathcal{O}$.
- $\mathcal{B}:\{\mathcal{C}, \hat{x}, \hat{y}, \hat{z}\}$: **Body-fixed reference frame**
A frame fixed in the rigid body with its origin at the body's center of mass $\mathcal{C}$, rotating with respect to $\mathcal{N}$ at $\omega_{\mathcal{B}/\mathcal{N}}$ (angular rate).
- $\mathcal{P}:\{\mathcal{O}, \hat{e}, \hat{p}, \hat{h}\}$: **Perifocal reference frame**
A frame fixed in the orbit plane with its origin at the attracting center $\mathcal{O}$, where $\hat{e}$ points toward periapsis along the eccentricity vector, $\hat{h}$ is along the angular momentum vector, and $\hat{p} = \hat{h} \times \hat{e}$ completes the right-handed set.

(h.1): Find

Derive the attitude equations of motion: Euler's equations with the gravity-gradient torque, together with a kinematic differential equation (KDEs) for my choice of attitude representation (DCM, Euler parameters, or Euler angles).

Analysis/Conclusions: It was already established in part (c) that the vector form of the Euler's equations of rotational motion can be represented as:

$$\boldsymbol{\tau} = I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \quad (59)$$

Where $\boldsymbol{\tau}$ is some torque applied on a rigid body, I_c is the inertia tensor of the rigid body about the center of mass, and $\boldsymbol{\omega}$ is the angular rate of the rigid body's rotation (which just like before is the same as $\omega_{\mathcal{B}/\mathcal{N}}$).

Likewise, since this differential equation provides a generic torque term, we can simply substitute in the vector equation of gravity gradient torque:

$$\boldsymbol{\tau}_G = \frac{3\mu}{r^5} \mathbf{r} \times (I_c \mathbf{r}) \quad (60)$$

Where μ is the gravitational parameter of the orbited body and $\mathbf{r}$ is the position vector of the rigid body with respect to the attractive center of the orbited body.

The substitution, and derivation of the Euler's equations including gravity gradient torque goes as follows:

$$\begin{aligned}\boldsymbol{\tau}_g &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\ \frac{3\mu}{r^5} \mathbf{r} \times (I_c \mathbf{r}) &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega})\end{aligned}\tag{61}$$

To simplify the terms in this expression, we can factor out the scalar of the $\mathbf{r}$ terms from its unit vector $\mathbf{r} = r \cdot \hat{\mathbf{r}}$:

$$\begin{aligned}\frac{3\mu}{r^5} (r^2) \hat{\mathbf{r}} \times (I_c \hat{\mathbf{r}}) &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\ \frac{3\mu}{r^3} \hat{\mathbf{r}} \times (I_c \hat{\mathbf{r}}) &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega})\end{aligned}\tag{62}$$

Using the definition of mean motion n , which is equal to $n = \sqrt{\mu/a^3}$ where a is the orbit semi-major axis, since this is a circular orbit ($e = 0$), the orbital radius is constant and equal to the semi-major axis ($r = a$). Thus the following substitution can be made in the above expression:

$$\begin{aligned}3 \frac{\mu}{a^3} \hat{\mathbf{r}} \times (I_c \hat{\mathbf{r}}) &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\ 3n^2 \hat{\mathbf{r}} \times (I_c \hat{\mathbf{r}}) &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega})\end{aligned}\tag{63}$$

Which collapses the μ and r dependence into the single constant n^2 . On an eccentric orbit r would vary with true anomaly and this coefficient would have to be recomputed at every integration step because the orbit is circular it is evaluated once. Then for the sake of numerical integration, the $\dot{\boldsymbol{\omega}}$ is isolated on one side of the equation:

$$\begin{aligned}3n^2 \hat{\mathbf{r}} \times (I_c \hat{\mathbf{r}}) &= I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\ I_c \dot{\boldsymbol{\omega}} &= 3n^2 \hat{\mathbf{r}} \times (I_c \hat{\mathbf{r}}) - \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) \\ \dot{\boldsymbol{\omega}} &= I_c^{-1} (3n^2 \hat{\mathbf{r}} \times (I_c \hat{\mathbf{r}}) - \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}))\end{aligned}\tag{64}$$

It should be noted that this equation is a vector relation, and thus it holds true in any reference frame we choose. However, for the sake of numerical integration it needs to be resolved into $\mathcal{B}$ -frame components, since I_c is only constant in the body frame. This means the unit position vector $\hat{\mathbf{r}}$ also needs to be expressed in body components. However, $\hat{\mathbf{r}}$ is not one of the integrated states, and is instead computed at each step from the orbit geometry and whatever the current attitude happens to be. Since we know this is a circular orbit, the true anomaly $\nu(t)$ simply advances linearly with time as a function of $n \cdot t$, so $\hat{\mathbf{r}}$ can be written in the $\mathcal{P}$ -frame as a unit circle on the $\hat{e} - \hat{p}$ plane, and then mapped

through $\mathcal{N}$ into $\mathcal{B}$:

$$\begin{aligned}
\nu(t) &= \nu_0 + nt \\
{}^{\mathcal{P}}\hat{\mathbf{r}} &= [\cos \nu(t), \sin \nu(t), 0]^T \\
{}^{\mathcal{N}}\hat{\mathbf{r}} &= [NP] {}^{\mathcal{P}}\hat{\mathbf{r}} \\
{}^{\mathcal{B}}\hat{\mathbf{r}} &= [BN(\mathbf{q})] {}^{\mathcal{N}}\hat{\mathbf{r}}
\end{aligned} \tag{65}$$

Where $[NP]$ is a DCM used in problem set 1 to map from the Perifocal frame to the $\mathcal{N}$ frame, evaluated at the periapsis direction, meaning the true longitude θ replaced by the argument of periapsis ω_0 . Note that this DCM is the transpose of a 3-1-3 Euler sequence rotated through Ω_0 , i , and ω_0 , since it maps out of the perifocal frame rather than into it:

$$[NP] = \begin{bmatrix} \cos \Omega_0 \cos \omega_0 - \sin \Omega_0 \cos i \sin \omega_0 & -\cos \Omega_0 \sin \omega_0 - \sin \Omega_0 \cos i \cos \omega_0 & \sin \Omega_0 \sin i \\ \sin \Omega_0 \cos \omega_0 + \cos \Omega_0 \cos i \sin \omega_0 & -\sin \Omega_0 \sin \omega_0 + \cos \Omega_0 \cos i \cos \omega_0 & -\cos \Omega_0 \sin i \\ \sin i \sin \omega_0 & \sin i \cos \omega_0 & \cos i \end{bmatrix} \tag{66}$$

Since the orbital elements are constant for an unperturbed circular orbit, $[NP]$ is built a single time from the values given in Table 1 rather than recomputed at each step.

And $[BN(\mathbf{q})]$ (given by the lecture slides) is the DCM mapping the ECI frame to the body frame, built from the current satellite attitude represented as an euler parameter, which is given as as:

$$[BN(\mathbf{q})] = \begin{bmatrix} 1 - 2q_2^2 - 2q_3^2 & 2(q_1q_2 + q_3q_4) & 2(q_1q_3 - q_2q_4) \\ 2(q_1q_2 - q_3q_4) & 1 - 2q_1^2 - 2q_3^2 & 2(q_2q_3 + q_1q_4) \\ 2(q_1q_3 + q_2q_4) & 2(q_2q_3 - q_1q_4) & 1 - 2q_1^2 - 2q_2^2 \end{bmatrix} \tag{67}$$

Unlike $[NP]$, this DCM must be rebuilt at every integration step, since $\mathbf{q}$ is one of the propagated states. Likewise, we must define the KDE to propagate the euler parameters over time as a function of angular rate $\boldsymbol{\omega}$ which is also given by the lecture slides as:

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \mathbf{q} \tag{68}$$

Where $\boldsymbol{\omega} = [\omega_1, \omega_2, \omega_3]^T$ is propagated from the gravity-gradient Euler's EoM, and $\mathbf{q} = [q_1, q_2, q_3, q_4]^T$.

Then the complete set of rotational EoM's and KDE's can be listed as:

$$\begin{aligned}\dot{\boldsymbol{\omega}} &= I_c^{-1} (3n^2 {}^{\mathcal{B}}\hat{\mathbf{r}} \times (I_c {}^{\mathcal{B}}\hat{\mathbf{r}}) - \boldsymbol{\omega} \times (I_c \boldsymbol{\omega})) \\ \dot{\mathbf{q}} &= \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \mathbf{q}\end{aligned}\quad (69)$$

$${}^{\mathcal{B}}\hat{\mathbf{r}} = [BN(\mathbf{q})] [NP] [\cos \nu(t), \sin \nu(t), 0]^T, \quad \nu(t) = \nu_0 + nt$$

$$[BN(\mathbf{q})] = \begin{bmatrix} 1 - 2q_2^2 - 2q_3^2 & 2(q_1q_2 + q_3q_4) & 2(q_1q_3 - q_2q_4) \\ 2(q_1q_2 - q_3q_4) & 1 - 2q_1^2 - 2q_3^2 & 2(q_2q_3 + q_1q_4) \\ 2(q_1q_3 + q_2q_4) & 2(q_2q_3 - q_1q_4) & 1 - 2q_1^2 - 2q_2^2 \end{bmatrix}\quad (70)$$

$$[NP] = \begin{bmatrix} \cos \Omega_0 \cos \omega_0 - \sin \Omega_0 \cos i \sin \omega_0 & -\cos \Omega_0 \sin \omega_0 - \sin \Omega_0 \cos i \cos \omega_0 & \sin \Omega_0 \sin i \\ \sin \Omega_0 \cos \omega_0 + \cos \Omega_0 \cos i \sin \omega_0 & -\sin \Omega_0 \sin \omega_0 + \cos \Omega_0 \cos i \cos \omega_0 & -\cos \Omega_0 \sin i \\ \sin i \sin \omega_0 & \sin i \cos \omega_0 & \cos i \end{bmatrix}\quad (71)$$

Where $[NP]$ is constant and $[BN(\mathbf{q})]$ is rebuilt from $\mathbf{q}$ at each step, making this a coupled seven state system $\mathbf{x} = [\mathbf{q}^T, \boldsymbol{\omega}^T]^T$.

(h.2): Find

Choose the initial attitude to be "nadir-pointing," where one body axis is aligned with $-\mathbf{r}_c$, while another aligned with the orbit-normal direction $\mathbf{r}_c \times \mathbf{v}_c$. There is some freedom in how I define "nadir-pointing"; state my convention and report the attitude initial condition.

Analysis/Conclusions:

For this part I define "nadir-pointing" as the $\mathcal{B}$ -frame vector $\hat{\mathbf{y}}$ being aligned with $-\mathbf{r}_c$ while the $\hat{\mathbf{x}}$ vector is aligned with $\mathbf{r}_c \times \mathbf{v}_c$.

To report the attitude initial condition (which would be the attitude in $\mathcal{N}$ -frame), I need to first define $\hat{\mathbf{r}}$ in the same frame.

We know from the initial conditions given in table 1, that the true anomaly starts at 0 deg which means it starts at periapsis. This also means that $\hat{\mathbf{r}}$ starts aligned with the $\hat{\mathbf{e}}$ vector in the $\mathcal{P}$ -frame and we can use the $[NP]$ DCM defined in the previous part:

$$\begin{aligned}[NP] &= \begin{bmatrix} \cos \Omega_0 \cos \omega_0 - \sin \Omega_0 \cos i \sin \omega_0 & -\cos \Omega_0 \sin \omega_0 - \sin \Omega_0 \cos i \cos \omega_0 & \sin \Omega_0 \sin i \\ \sin \Omega_0 \cos \omega_0 + \cos \Omega_0 \cos i \sin \omega_0 & -\sin \Omega_0 \sin \omega_0 + \cos \Omega_0 \cos i \cos \omega_0 & -\cos \Omega_0 \sin i \\ \sin i \sin \omega_0 & \sin i \cos \omega_0 & \cos i \end{bmatrix} \\ &= \begin{bmatrix} \cos 0 \cos 0 - \sin 0 \cos 45 \sin 0 & -\cos 0 \sin 0 - \sin 0 \cos 45 \cos 0 & \sin 0 \sin 45 \\ \sin 0 \cos 0 + \cos 0 \cos 45 \sin 0 & -\sin 0 \sin 0 + \cos 0 \cos 45 \cos 0 & -\cos 0 \sin 45 \\ \sin 45 \sin 0 & \sin 45 \cos 0 & \cos 45 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.7071\dots & -0.7071\dots \\ 0 & 0.7071\dots & 0.7071\dots \end{bmatrix}\end{aligned}\quad (72)$$

So, using this DCM, we can simply take the $\mathcal{P}$ -frame unit vector at periapsis (defined as ${}^{\mathcal{P}}\hat{\mathbf{r}} = [1, 0, 0]^t$) and transform it to $\mathcal{N}$ -frame. Recall we established

$\hat{\mathbf{e}} = {}^{\mathcal{P}}\hat{\mathbf{r}}$ at epoch:

$$\begin{aligned}
{}^{\mathcal{N}}\hat{\mathbf{r}} &= [NP] {}^{\mathcal{P}}\hat{\mathbf{r}} \\
{}^{\mathcal{N}}\hat{\mathbf{r}} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.7071... & -0.7071... \\ 0 & 0.7071... & 0.7071... \end{bmatrix} \hat{\mathbf{e}} \\
\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.7071... & -0.7071... \\ 0 & 0.7071... & 0.7071... \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\
{}^{\mathcal{N}}\hat{\mathbf{r}} &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \hat{\mathbf{e}} = {}^{\mathcal{P}}\hat{\mathbf{r}}
\end{aligned} \tag{73}$$

So, with the $\mathcal{N}$ -frame value of $\hat{\mathbf{r}}$ defined, we also need to understand the alignment of the rigid body with $\hat{\mathbf{r}} \times \hat{\mathbf{v}}$, which would also be equivalent to the specific angular momentum unit vector $\hat{\mathbf{h}}$. This requires the velocity direction at epoch. Since the orbit is circular, the flight path angle is zero everywhere, meaning the velocity is always purely transverse and therefore lies along the $\hat{\mathbf{p}}$ basis vector of the $\mathcal{P}$ -frame at periapsis. Mapping it into the $\mathcal{N}$ -frame the same way as before:

$$\begin{aligned}
{}^{\mathcal{N}}\hat{\mathbf{v}} &= [NP] {}^{\mathcal{P}}\hat{\mathbf{v}} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.7071... & -0.7071... \\ 0 & 0.7071... & 0.7071... \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.7071... \\ 0.7071... \end{bmatrix}
\end{aligned} \tag{74}$$

Then the orbit-normal direction follows from the cross product of the two:

$$\begin{aligned}
{}^{\mathcal{N}}\hat{\mathbf{h}} &= {}^{\mathcal{N}}\hat{\mathbf{r}} \times {}^{\mathcal{N}}\hat{\mathbf{v}} \\
&= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0.7071... \\ 0.7071... \end{bmatrix} = \begin{bmatrix} 0 \\ -0.7071... \\ 0.7071... \end{bmatrix}
\end{aligned} \tag{75}$$

With both named directions now expressed in the $\mathcal{N}$ -frame, the orientation of the $\mathcal{B}$ -frame at epoch follows directly from the convention. The $\hat{\mathbf{y}}$ axis points along $-\hat{\mathbf{r}}$, the $\hat{\mathbf{x}}$ axis points along $\hat{\mathbf{h}}$, and the $\hat{\mathbf{z}}$ axis is whatever completes the right handed set:

$$\begin{aligned}
{}^{\mathcal{N}}\hat{\mathbf{x}} &= {}^{\mathcal{N}}\hat{\mathbf{h}} = [0, -0.7071..., 0.7071...]^T \\
{}^{\mathcal{N}}\hat{\mathbf{y}} &= -{}^{\mathcal{N}}\hat{\mathbf{r}} = [-1, 0, 0]^T \\
{}^{\mathcal{N}}\hat{\mathbf{z}} &= {}^{\mathcal{N}}\hat{\mathbf{x}} \times {}^{\mathcal{N}}\hat{\mathbf{y}} = [0, -0.7071..., -0.7071...]^T
\end{aligned} \tag{76}$$

These three unit vectors are the $\mathcal{B}$ -frame basis expressed in $\mathcal{N}$ -frame components, which is exactly the definition of the rows of the DCM $[BN]$. So the attitude at epoch requires no further computation and can simply be assembled

by stacking them:

$$[BN(t_0)] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{x}}^T \\ \mathcal{N} \hat{\mathbf{y}}^T \\ \mathcal{N} \hat{\mathbf{z}}^T \end{bmatrix} = \begin{bmatrix} 0 & -0.7071... & 0.7071... \\ -1 & 0 & 0 \\ 0 & -0.7071... & -0.7071... \end{bmatrix} \quad (77)$$

Since the KDE derived in part (h.1) propagates Euler parameters rather than a DCM, this attitude must be converted into that representation. A method to directly compute the Euler parameter components from a DCM like $[BN]$ is given by [1]:

$$\begin{aligned} q_4 &= \pm \frac{1}{2} \sqrt{\text{tr}[BN] + 1}, & q_1 &= \frac{BN_{23} - BN_{32}}{4q_4} \\ q_2 &= \frac{BN_{31} - BN_{13}}{4q_4}, & q_3 &= \frac{BN_{12} - BN_{21}}{4q_4} \end{aligned} \quad (78)$$

The positive root is taken here for q_4 . The trace of $[BN(t_0)]$ is $0+0-0.7071... = -0.7071...$, so the scalar parameter is:

$$q_4 = \frac{1}{2} \sqrt{-0.7071... + 1} = \frac{1}{2} \sqrt{0.2928...} = 0.2705... \quad (79)$$

And with $4q_4 = 1.0823...$, the three vector parameters are:

$$\begin{aligned} q_1 &= \frac{0 - (-0.7071...)}{1.0823...} = \frac{0.7071...}{1.0823...} = 0.6532... \\ q_2 &= \frac{0 - 0.7071...}{1.0823...} = \frac{-0.7071...}{1.0823...} = -0.6532... \\ q_3 &= \frac{-0.7071... - (-1)}{1.0823...} = \frac{0.2928...}{1.0823...} = 0.2705... \end{aligned} \quad (80)$$

Which gives the attitude initial condition as the Euler parameter set:

$$\mathbf{q}(t_0) = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \begin{bmatrix} 0.6532... \\ -0.6532... \\ 0.2705... \\ 0.2705... \end{bmatrix} \quad (81)$$

(h.3): Find

Perturb your initial attitude by a small offset (a few degrees) with a small initial body rate (report the values), integrate the attitude motion for several orbits, and plot the resulting attitude/rate time history. Describe what you observe empirically (does the small perturbation stay bounded, grow, oscillate?)

Analysis/Conclusions: To perturb the initial attitude $\mathbf{q}_0$ by a few degrees, we can simply take some angular rate equal to about 3 deg/s and numerically

integrate the Euler parameter for 1 second using the KDE defined for Euler parameter propagation in (h.3).

So using a perturbation by integrating in the method described above using:

$$\boldsymbol{\omega}_{\text{pert}} = [3 \text{ deg}/s, 0, 0]^T = [0.0523... \text{rad}/s, 0, 0]^T \quad (82)$$

We get an initial perturbed attitude as an Euler parameter of:

$$\mathbf{q}(t_0) = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \begin{bmatrix} 0.6601... \\ -0.6459... \\ 0.2876... \\ 0.2534... \end{bmatrix} \quad (83)$$

And for this simulation I use an initial body rate with each axis at positive $0.01 \text{ deg}/s$:

$$\boldsymbol{\omega}_{\text{pert}} = \begin{bmatrix} 1.74532... \times 10^{-4} \text{rad}/s \\ 1.74532... \times 10^{-4} \text{rad}/s \\ 1.74532... \times 10^{-4} \text{rad}/s \end{bmatrix} \quad (84)$$

The time span for this simulation was requested for several orbits, which I arbitrarily picked to be 5 orbits. The orbital period can be found by:

$$\begin{aligned} P &= 2\pi \sqrt{\frac{a^3}{\mu}} \\ &= 2\pi \sqrt{\frac{(6978 \text{ km})^3}{398600 \text{ km}^3/s^2}} \\ &= 5,801.0641...s \left(\frac{1 \text{ min}}{60s} \right) \left(\frac{1 \text{ hr}}{60 \text{ min}} \right) = 1.6114... \text{ hr} \\ t_f &= 5 \cdot P = 5 \cdot 5,801.0641 \approx 29,000s \end{aligned} \quad (85)$$

So the simulation will run for $29,000s$ to get approximately 5 orbits of data.

After running the requested simulation, the following is the requested attitude time history represented as Euler parameters:

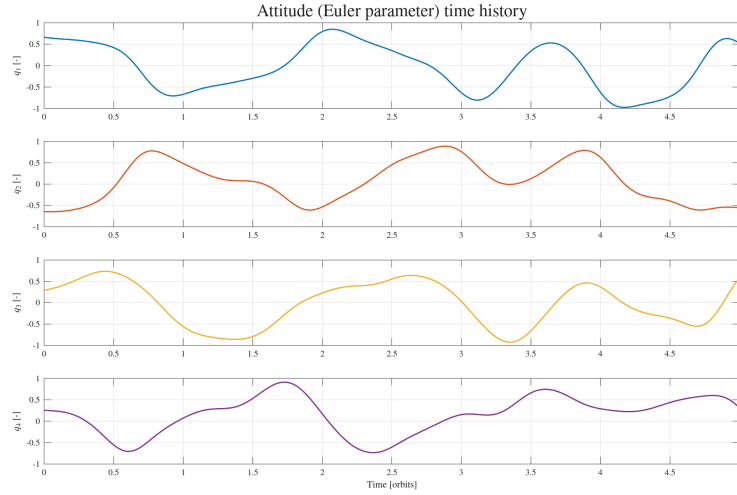


Figure 5: Time history of Euler parameters over five orbits.

And the time history of the angular rate:

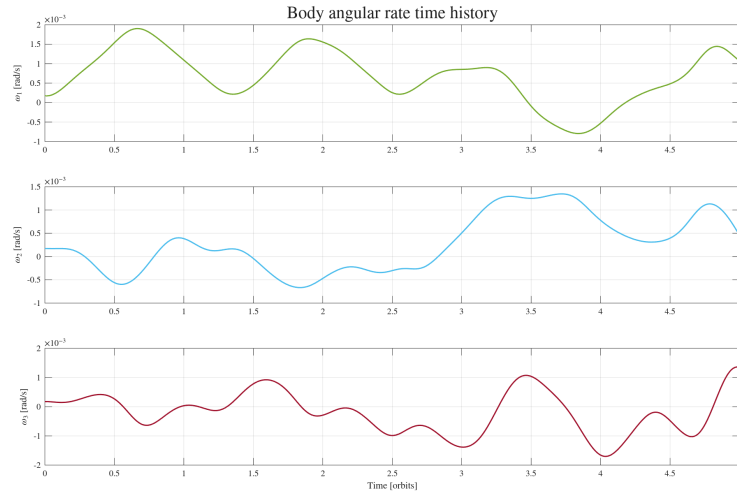


Figure 6: Time history of angular rate over five orbits.

From the observations of the plots above, we see that despite fluctuation in both the attitude and angular rate, the initial perturbation state combining angular rate and attitude remains bounded. There is no shrinkage or growth that seem to fit any distinguishable pattern.

Likewise, the four states of the Euler parameter seems to not relate to each other in any meaningful way and they appear to take somewhat of a random walk across the time history. The order of magnitude of these histories stay consistent which is expected because the Euler parameter is constrained to be unit length.

The angular rate plot follows a similar behavior where there is no noticeable relation between the changes of the three. The order of magnitude of the three angular rates stays consistent and they are within the same range of each other on the three axes.

References

- [1] H. Schaub and J. L. Junkins, *Analytical Mechanics of Space Systems, Fourth Edition*. Reston, VA: American Institute of Aeronautics and Astronautics, Inc., Apr. 2018.